

upstream d_1 applies T per client

(P1) value: $S(T(U)) = S(U)$

(P2) ordering: S 's induced ranking is fixed

(P3) decision: d_2 decides the same way

(P1) $\Rightarrow$ (P2) $\Rightarrow$ (P3): implications, true by definition

IT BREAKS HERE, IN BOTH DIRECTIONS

(P3) held, (P5) failed: cos_krum's selection is bit-identical at every rung (decision, admission 0.000), yet ASR falls 0.425

(P4) admission: the adversarial mass d_2 admits is unchanged

(P3) failed, (P5) held: Krum's decision flips 0.80 of rounds with admission $\equiv$ 0.000 (adversarial share pinned at 0.267), and ASR stays flat (0.016–0.062)

(P5) suppression: ASR stays low

**Statistic preservation is not attack-suppression preservation:
only the (P4) $\rightarrow$ (P5) link carries the security content in our experiments.**